Compatibility Relation for Point Groups

Table 1: $O_{\rm h}$ subgroup.

$O_{\rm h}$	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$T_{1g}(x)$	$T_{1g}(y)$	$T_{1g}(z)$	$T_{2g}(x)$	$T_{2g}(y)$	$T_{2g}(z)$	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$T_{1u}(x)$	$T_{1u}(y)$	$T_{1u}(z)$	$T_{2u}(x)$	$T_{2u}(y)$	$T_{2u}(z)$
Q	Q_0/Q_4	Q_{63}	Q_u/Q_{4u}	Q_v/Q_{4v}	Q^{α}_{4x}	Q^{α}_{4y}	Q^{α}_{4z}	Q_{yz}/Q^{β}_{4x}	Q_{xz}/Q^{β}_{4y}	$Q_{xy}/Q^{\beta}_{4z}/Q_{6xy}$	Q_{9xyz}	Q_{xyz}	Q_{5u}	Q_{5v}	Q_x/Q^{α}_x	Q_y/Q^{α}_y	Q_z/Q^{α}_z	Q^{β}_x	Q^{β}_y	Q^{β}_z
G	G_{9xyz}	G_{xyz}	G_{5u}	G_{5v}	G_x/G^{α}_x	G_y/G^{α}_y	G_z/G^{α}_z	G^{β}_x	G^{β}_y	G^{β}_z	G_0/G_4	G_{63}	G_u/G_{4u}	G_v/G_{4v}	G^{α}_{4x}	G^{α}_{4y}	G^{α}_{4z}	G_{yz}/G^{β}_{4x}	G_{xz}/G^{β}_{4y}	$G_{xy}/G^{\beta}_{4z}/G_{6xy}$
$O_{\rm h}$	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$T_{1g}(x)$	$T_{1g}(y)$	$T_{1g}(z)$	$T_{2g}(x)$	$T_{2g}(y)$	$T_{2g}(z)$	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$T_{1u}(x)$	$T_{1u}(y)$	$T_{1u}(z)$	$T_{2u}(x)$	$T_{2u}(y)$	$T_{2u}(z)$
$T_{\rm d}$	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	A_2	A_1	$E(y)$	$-E(x)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	$T_1(x)$	$T_1(y)$	$T_1(z)$
O	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$
$T_{\rm h}$	A_g	A_g	$E_g(x)$	$E_g(y)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	A_u	A_u	$E_u(x)$	$E_u(y)$	$T_u(x)$	$T_u(y)$	$T_u(z)$	$T_u(x)$	$T_u(y)$	$T_u(z)$
$T_{\rm h}$(c)	A_g	A_g	$E^{(a)}_g$	$E^{(b)}_g$	$T_g(x)$	$T_g(y)$	$T_g(z)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	A_u	A_u	$E^{(a)}_u$	$E^{(b)}_u$	$T_u(x)$	$T_u(y)$	$T_u(z)$	$T_u(x)$	$T_u(y)$	$T_u(z)$
T	A	A	$E(x)$	$E(y)$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$	A	A	$E(x)$	$E(y)$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$
T(c)	A	A	$E^{(a)}$	$E^{(b)}$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$	A	A	$E^{(a)}$	$E^{(b)}$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$
$D_{4\rm h}$	A_{1g}	B_{1g}	A_{1g}	B_{1g}	$E_g(x)$	$-E_g(y)$	A_{2g}	$E_g(x)$	$E_g(y)$	B_{2g}	A_{1u}	B_{1u}	A_{1u}	B_{1u}	$E_u(x)$	$E_u(y)$	A_{2u}	$E_u(x)$	$-E_u(y)$	B_{2u}
$D_{2\rm d}$	A_1	B_1	A_1	B_1	$E(x)$	$-E(y)$	A_2	$E(x)$	$E(y)$	B_2	B_1	A_1	B_1	A_1	$E(x)$	$E(y)$	B_2	$E(x)$	$-E(y)$	A_2
$D_{2\rm d}(1)$	A_1	B_2	A_1	B_2	$E(y)$	$E(x)$	A_2	$E(y)$	$-E(x)$	B_1	B_1	A_2	B_1	A_2	$E(x)$	$E(y)$	B_2	$E(x)$	$-E(y)$	A_1
$C_{4\rm v}$	A_1	B_1	A_1	B_1	$E(y)$	$-E(x)$	A_2	$E(y)$	$E(x)$	B_2	A_2	B_2	A_2	B_2	$E(x)$	$E(y)$	A_1	$E(x)$	$-E(y)$	B_1
D_4	A_1	B_1	A_1	B_1	$E(x)$	$E(y)$	A_2	$E(x)$	$-E(y)$	B_2	A_1	B_1	A_1	B_1	$E(x)$	$E(y)$	A_2	$E(x)$	$-E(y)$	B_2
$C_{4\rm h}$	A_g	B_g	A_g	B_g	$E_g(y)$	$E_g(x)$	A_g	$E_g(x)$	$E_g(y)$	B_g	A_u	B_u	A_u	B_u	$E_u(x)$	$E_u(y)$	A_u	$E_u(y)$	$E_u(x)$	B_u
$C_{4\rm h}$(c)	A_g	B_g	A_g	B_g	$E^{(a)}_g$	$E^{(b)}_g$	A_g	$E^{(b)}_g$	$E^{(a)}_g$	B_g	A_u	B_u	A_u	B_u	$E^{(a)}_u$	$E^{(b)}_u$	A_u	$E^{(b)}_u$	$E^{(a)}_u$	B_u
S_4	A	B	A	B	$E(y)$	$E(x)$	A	$E(x)$	$E(y)$	B	B	A	B	A	$E(x)$	$E(y)$	B	$E(y)$	$E(x)$	A
S_4(c)	A	B	A	B	$E^{(b)}$	$E^{(a)}$	A	$E^{(a)}$	$E^{(b)}$	B	B	A	B	A	$E^{(a)}$	$E^{(b)}$	B	$E^{(b)}$	$E^{(a)}$	A
C_4	A	B	A	B	$E(x)$	$E(y)$	A	$E(y)$	$E(x)$	B	A	B	A	B	$E(x)$	$E(y)$	A	$E(y)$	$E(x)$	B
C_4(c)	A	B	A	B	$E^{(a)}$	$E^{(b)}$	A	$E^{(b)}$	$E^{(a)}$	B	A	B	A	B	$E^{(a)}$	$E^{(b)}$	A	$E^{(b)}$	$E^{(a)}$	B
$D_{2\rm h}$	A_g	A_g	A_g	A_g	B_{3g}	B_{2g}	B_{1g}	B_{3g}	B_{2g}	B_{1g}	A_u	A_u	A_u	A_u	B_{3u}	B_{2u}	B_{1u}	B_{3u}	B_{2u}	B_{1u}
$C_{2\rm v}$	A_1	A_1	A_1	A_1	B_2	B_1	A_2	B_2	B_1	A_2	A_2	A_2	A_2	A_2	B_1	B_2	A_1	B_1	B_2	A_1
D_2	A	A	A	A	B_3	B_2	B_1	B_3	B_2	B_1	A	A	A	A	B_3	B_2	B_1	B_3	B_2	B_1
$C_{2\rm h}$	A_g	A_g	A_g	A_g	B_g	A_g	B_g	B_g	A_g	B_g	A_u	A_u	A_u	A_u	B_u	A_u	B_u	B_u	A_u	B_u
$C_{\rm s}$	A'	A'	A'	A'	A''	A'	A''	A''	A'	A''	A''	A''	A''	A''	A'	A''	A'	A'	A''	A'
C_2	A	A	A	A	B	A	B	B	A	B	A	A	A	A	B	A	B	B	A	B
$C_{\rm i}$	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u
C_1	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A

Table 2: $D_{6\mathrm{h}}$ subgroup.

$D_{6\mathrm{h}}$	A_{1g}	A_{2g}	B_{1g}	B_{2g}	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_{1u}	A_{2u}	B_{1u}	B_{2u}	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
Q	$Q_0/Q_u/Q_{40}$	Q_{6xy}	Q_{4c}	Q_{4s}	Q_{yz}/Q^{α}_{4u}	$-Q_{xz}/Q^{\alpha}_{4v}$	$Q_v/Q_{4c4}/Q_{4v}$	$-Q_{xy}/Q^{\alpha}_{4z}/Q^{\beta}_{4z}$	Q_{7xyz}	Q_z/Q^{α}_z	Q_{3s}	Q_{3c}	Q_x/Q_{3x}	Q_y/Q_{3y}	$Q_{xyz}/-Q_{5u}/-Q_{5v}$	Q^{β}_z
G	G_{7xyz}	G_z/G^{α}_z	G_{3s}	G_{3c}	G_x/G_{3x}	G_y/G_{3y}	$G_{xyz}/-G_{5u}/-G_{5v}$	G^{β}_z	$G_0/G_u/G_{40}$	G_{6xy}	G_{4c}	G_{4s}	G_{yz}/G^{α}_{4u}	$-G_{xz}/G^{\alpha}_{4v}$	$G_v/G_{4c4}/G_{4v}$	$-G_{xy}/G^{\alpha}_{4z}/G^{\beta}_{4z}$
$D_{6\mathrm{h}}$	A_{1g}	A_{2g}	B_{1g}	B_{2g}	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_{1u}	A_{2u}	B_{1u}	B_{2u}	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
$D_{3\mathrm{h}}$	A'_1	A'_2	A''_1	A''_2	$E''(x)$	$E''(y)$	$E'(y)$	$-E'(x)$	A''_1	A''_2	A'_1	A'_2	$E'(x)$	$E'(y)$	$E''(y)$	$-E''(x)$
$D_{3\mathrm{h}}(1)$	A'_1	A'_2	A''_2	A''_1	$E''(x)$	$E''(y)$	$E'(x)$	$E'(y)$	A''_1	A''_2	A'_2	A'_1	$E'(x)$	$E'(y)$	$E''(x)$	$E''(y)$
$C_{6\mathrm{v}}$	A_1	A_2	B_2	B_1	$E_1(y)$	$-E_1(x)$	$E_2(x)$	$E_2(y)$	A_2	A_1	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(y)$	$-E_2(x)$
D_6	A_1	A_2	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$	A_1	A_2	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$
$C_{6\mathrm{h}}$	A_g	A_g	B_g	B_g	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_u	A_u	B_u	B_u	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
$C_{6\mathrm{h}}(\mathrm{c})$	A_g	A_g	B_g	B_g	$E^{(a)}_{1g}$	$E^{(b)}_{1g}$	$E^{(a)}_{2g}$	$E^{(b)}_{2g}$	A_u	A_u	B_u	B_u	$E^{(a)}_{1u}$	$E^{(b)}_{1u}$	$E^{(a)}_{2u}$	$E^{(b)}_{2u}$
$C_{3\mathrm{h}}$	A'	A'	A''	A''	$E''(x)$	$E''(y)$	$E'(x)$	$E'(y)$	A''	A''	A'	A'	$E'(x)$	$E'(y)$	$E''(x)$	$E''(y)$
$C_{3\mathrm{h}}(\mathrm{c})$	A'	A'	A''	A''	$E''^{(a)}$	$E''^{(b)}$	$E'^{(a)}$	$E'^{(b)}$	A''	A''	A'	A'	$E'^{(a)}$	$E'^{(b)}$	$E''^{(a)}$	$E''^{(b)}$
C_6	A	A	B	B	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$	A	A	B	B	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$
$C_6(\mathrm{c})$	A	A	B	B	$E^{(a)}_1$	$E^{(b)}_1$	$E^{(a)}_2$	$E^{(b)}_2$	A	A	B	B	$E^{(a)}_1$	$E^{(b)}_1$	$E^{(a)}_2$	$E^{(b)}_2$
$D_{3\mathrm{d}}$	A_{1g}	A_{2g}	A_{2g}	A_{1g}	$E_g(x)$	$E_g(y)$	$E_g(x)$	$E_g(y)$	A_{1u}	A_{2u}	A_{2u}	A_{1u}	$E_u(x)$	$E_u(y)$	$E_u(x)$	$E_u(y)$
$D_{3\mathrm{d}}(1)$	A_{1g}	A_{2g}	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$E_g(y)$	$-E_g(x)$	A_{1u}	A_{2u}	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$E_u(y)$	$-E_u(x)$
$C_{3\mathrm{v}}$	A_1	A_2	A_2	A_1	$E(y)$	$-E(x)$	$E(y)$	$-E(x)$	A_2	A_1	A_1	A_2	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$C_{3\mathrm{v}}(1)$	A_1	A_2	A_1	A_2	$E(y)$	$-E(x)$	$E(x)$	$E(y)$	A_2	A_1	A_2	A_1	$E(x)$	$E(y)$	$E(y)$	$-E(x)$
D_3	A_1	A_2	A_2	A_1	$E(x)$	$E(y)$	$E(x)$	$E(y)$	A_1	A_2	A_2	A_1	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$D_3(1)$	A_1	A_2	A_1	A_2	$E(x)$	$E(y)$	$E(y)$	$-E(x)$	A_1	A_2	A_1	A_2	$E(x)$	$E(y)$	$E(y)$	$-E(x)$
$C_{3\mathrm{i}}$	A_g	A_g	A_g	A_g	$E_g(x)$	$E_g(y)$	$E_g(x)$	$E_g(y)$	A_u	A_u	A_u	A_u	$E_u(x)$	$E_u(y)$	$E_u(x)$	$E_u(y)$
$C_{3\mathrm{i}}(\mathrm{c})$	A_g	A_g	A_g	A_g	$E^{(a)}_g$	$E^{(b)}_g$	$E^{(a)}_g$	$E^{(b)}_g$	A_u	A_u	A_u	A_u	$E^{(a)}_u$	$E^{(b)}_u$	$E^{(a)}_u$	$E^{(b)}_u$
C_3	A	A	A	A	$E(x)$	$E(y)$	$E(x)$	$E(y)$	A	A	A	A	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$C_3(\mathrm{c})$	A	A	A	A	$E^{(a)}$	$E^{(b)}$	$E^{(a)}$	$E^{(b)}$	A	A	A	A	$E^{(a)}$	$E^{(b)}$	$E^{(a)}$	$E^{(b)}$

Table 3: Orbitals (harmonics).

rank	tag	symbol	expression
0	`s`	$\mathbb{X}_0$	1
1	`px`	$\mathbb{X}_x$	x
	`py`	$\mathbb{X}_y$	y
	`pz`	$\mathbb{X}_z$	z
2	`du`	$\mathbb{X}_u$	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	`dv`	$\mathbb{X}_v$	$\frac{\sqrt{3}\left(x^2-y^2\right)}{2}$
	`dyz`	$\mathbb{X}_{yz}$	$\sqrt{3}yz$
	`dxz`	$\mathbb{X}_{xz}$	$\sqrt{3}xz$
	`dxy`	$\mathbb{X}_{xy}$	$\sqrt{3}xy$
3	`f3`	$\mathbb{X}_{xyz}$	$\sqrt{15}xyz$
	`faz`	$\mathbb{X}_z^{\alpha}$	$\frac{z\left(-3x^2-3y^2+2z^2\right)}{2}$
	`fbz`	$\mathbb{X}_z^{\beta}$	$\frac{\sqrt{15}z\left(x^2-y^2\right)}{2}$
	`fax`	$\mathbb{X}_x^{\alpha}$	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
	`fay`	$\mathbb{X}_y^{\alpha}$	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
	`fbx`	$\mathbb{X}_x^{\beta}$	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
	`fby`	$\mathbb{X}_y^{\beta}$	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
	`fc`	$\mathbb{X}_{3c}$	$\frac{\sqrt{10}x\left(x^2-3y^2\right)}{4}$
	`fs`	$\mathbb{X}_{3s}$	$\frac{\sqrt{10}y\left(3x^2-y^2\right)}{4}$
	`fx`	$\mathbb{X}_{3x}$	$-\frac{\sqrt{6}x\left(x^2+y^2-4z^2\right)}{4}$
	`fy`	$\mathbb{X}_{3y}$	$-\frac{\sqrt{6}y\left(x^2+y^2-4z^2\right)}{4}$
4	`g`	$\mathbb{X}_4$	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
	`gu`	$\mathbb{X}_{4u}$	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$

continued ...

Table 3

rank	tag	symbol	expression
	gv	$\mathbb{X}_{4v}$	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
	gax	$\mathbb{X}^{\alpha}_{4x}$	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
	gay	$\mathbb{X}^{\alpha}_{4y}$	$-\frac{\sqrt{35}xz(x-z)(x+z)}{2}$
	gaz	$\mathbb{X}^{\alpha}_{4z}$	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
	gbx	$\mathbb{X}^{\beta}_{4x}$	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
	gby	$\mathbb{X}^{\beta}_{4y}$	$-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$
	gbz	$\mathbb{X}^{\beta}_{4z}$	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
	g0	$\mathbb{X}_{40}$	$\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$
	gc	$\mathbb{X}_{4c}$	$\frac{\sqrt{70}xz\left(x^2-3y^2\right)}{4}$
	gs	$\mathbb{X}_{4s}$	$\frac{\sqrt{70}yz\left(3x^2-y^2\right)}{4}$
	gc4	$\mathbb{X}_{4c4}$	$\frac{\sqrt{35}\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$
	gau	$\mathbb{X}^{\alpha}_{4u}$	$-\frac{\sqrt{10}yz\left(3x^2+3y^2-4z^2\right)}{4}$
	gav	$\mathbb{X}^{\alpha}_{4v}$	$\frac{\sqrt{10}xz\left(3x^2+3y^2-4z^2\right)}{4}$
5	hu	$\mathbb{X}_{5u}$	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$
	hv	$\mathbb{X}_{5v}$	$\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$
6	i3	$\mathbb{X}_{63}$	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$
	ixy	$\mathbb{X}_{6xy}$	$\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$
7	j3	$\mathbb{X}_{7xyz}$	$\frac{\sqrt{6006}xyz\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$
9	l3	$\mathbb{X}_{9xyz}$	$-\frac{\sqrt{510510}xyz(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$